

AI Based Forecast Bust Detection

Feature wise implementation workplan for the SIH demo

Project Goal

Build a simple working prototype that estimates where an NWP forecast is likely to be unreliable, shows a regional forecast bust probability and confidence level, retrieves similar historical situations, and explains the result in plain meteorological language. The first demo should avoid unnecessary model training and use a straightforward hybrid approach: historical error analysis, analogue retrieval, statistical scoring, and AI explanation.

1. Forecast Data Layer

Priority	What needs to be done	How it would be done
High	Create the input layer for current and historical NWP forecast data.	For the demo, prepare a clean local dataset containing date, forecast cycle, lead time, region or grid, forecast variable, and forecast value. Start with rainfall and a limited geographic scope. Keep the data format stable so a real NWP API can replace the mock data later.

2. Observation Data Layer

Priority	What needs to be done	How it would be done
High	Provide the actual weather values needed to determine whether historical forecasts were wrong.	Prepare matching observations for the same timestamps, regions, and variables as the forecasts. Initially use a prepared CSV or JSON dataset. Later connect IMD or another suitable observation source.

3. Forecast and Observation Alignment

Priority	What needs to be done	How it would be done
High	Make forecast and actual observations directly comparable.	Match by valid time, forecast cycle, lead time, spatial grid or region, and variable. Handle missing values and make sure the system never uses future information when estimating a forecast issued at an earlier time.

4. Historical Error Engine

Priority	What needs to be done	How it would be done
High	Calculate how large the forecast errors were historically.	For every forecast and region, calculate absolute error, relative error where meaningful, MAE, RMSE, bias, and lead time specific error. Store these results as the historical error dataset used by the confidence engine.

5. Forecast Bust Definition

Priority	What needs to be done	How it would be done
High	Define what counts as a forecast bust.	Use a simple statistical threshold for the demo, preferably a regional historical percentile such as P90 rather than one fixed rainfall threshold. Label each historical case as normal or forecast bust.

6. Weather Fingerprint

Priority	What needs to be done	How it would be done
High	Convert a current forecast into a compact representation that can be compared with historical cases.	Combine available forecast variables and derived characteristics such as rainfall, pressure, humidity, wind, lead time, forecast revision, and selected geographic features. Normalize the values and create a consistent fingerprint.

7. Historical Analogue Retrieval

Priority	What needs to be done	How it would be done
High	Find past weather situations that resemble the current forecast.	Store historical weather fingerprints and retrieve the top similar cases using vector similarity or a simple numerical similarity score. Return the most similar events together with their forecast errors and bust labels.

8. Bust Probability and Confidence

Priority	What needs to be done	How it would be done
High	Convert historical evidence into an easy to understand confidence indicator.	Calculate the bust frequency among similar historical cases and combine it with regional historical error statistics and current forecast characteristics. Produce a probability from 0 to 1 and map it to High, Medium, or Low confidence. Keep the calculation deterministic and transparent for the demo.

9. India Confidence Map

Priority	What needs to be done	How it would be done
High	Show spatial forecast reliability across India.	Create a state or grid level map where each region receives a confidence value. Use green for high confidence, yellow for medium confidence, and red for low confidence. Allow Day 1 through Day 10 selection.

10. Regional Investigation

Priority	What needs to be done	How it would be done
High	Allow a user to inspect one region in detail.	When a state or region is clicked, display forecast value, bust probability, confidence, historical mean error, number of similar cases, historical bust frequency, and the main contributing factors.

11. Historical Analogue Panel

Priority	What needs to be done	How it would be done
Medium	Make the retrieved historical evidence visible to the user.	Show the top three to five similar historical events with date, region, event type, similarity, forecast error, and bust status. This provides evidence behind the confidence score.

12. AI Explanation

Priority	What needs to be done	How it would be done
High	Generate a concise explanation of why confidence is high or low.	Pass the calculated probability, current forecast features, historical error statistics, and retrieved analogue cases to an LLM. Instruct it to explain only the supplied evidence. The LLM should explain the result, not invent the probability.

13. Forecast Day and Variable Controls

Priority	What needs to be done	How it would be done
High	Allow the user to explore different forecast horizons and weather variables.	Add selectors for Day 1 through Day 10 and variables such as rainfall, temperature, wind, and pressure. For the first demo, rainfall can be the primary fully implemented variable while other variables use prepared mock values.

14. Forecast Confidence Dashboard

Priority	What needs to be done	How it would be done
High	Present the complete system in one operational interface.	Build a desktop first React interface with summary cards, India map, timeline, regional panel, analogue evidence, probability gauge, confidence indicator, AI explanation, and data source status.

15. Backend API Layer

Priority	What needs to be done	How it would be done
Medium	Connect the frontend to actual processing services.	Create FastAPI endpoints for forecast data, confidence maps, region details, historical analogues, and explanations. Replace local mock data with API calls without changing the frontend component structure.

16. Evaluation and Demo Case Study

Priority	What needs to be done	How it would be done
High	Prove that the concept works on historical situations.	Select one or two historical high error events. Run the system using information available before the event, show retrieved analogues, calculate the bust probability, and compare the prediction with the eventual observed error. Report simple metrics such as precision, recall, calibration, and detection lead time when enough data is available.

Recommended Build Order

1. Finish the frontend dashboard using deterministic mock data.
2. Create the historical forecast and observation sample dataset.
3. Implement forecast and observation alignment.
4. Implement historical error calculation and bust labels.
5. Build the weather fingerprint and historical analogue retrieval.
6. Implement the simple bust probability and confidence calculation.
7. Connect the confidence values to the India map and regional detail view.

8. Add the historical analogue evidence panel.
9. Add the LLM explanation using only quantitative evidence supplied by the system.
10. Create one complete historical case study and use it as the main SIH demo.

Minimum Viable Demo

Do not build everything at once. The minimum convincing demo is: rainfall forecast data, historical observations, forecast error calculation, a simple bust threshold, historical analogue retrieval, deterministic bust probability, an India confidence map, an Odisha or similar regional detail panel, and an AI explanation. The rest can be added after this end to end flow works.

Important Design Rule

The numerical system should calculate the bust probability. The LLM should explain that probability using retrieved evidence. Do not ask the LLM to invent a probability directly. This keeps the demo transparent, reproducible, and easier to defend technically.